

## DESIGN &amp; SETUP REPORT

# A transcranial focused-ultrasound targeting pipeline for the squirrel monkey: a *Saimiri sciureus* museum microCT registered to the VALiDATE29 brain atlas, with a hard CT-calibration guard (deliverables 1–2)

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**Abstract.** We document the fifth pillar of TUBA — a transcranial focused ultrasound (FUS) pipeline for the common squirrel monkey (*Saimiri sciureus*), prepared for NIH R01 EB037345 (Aim 3): per-animal skull models for time-reversal aberration correction and multifocal beam design targeting the hand representations of primary somatosensory (S1 areas 3b and 1) and primary motor (M1, area 4) cortex. This report covers **deliverables 1–2 (the geometry stages)** and is scoped, by design decision, to a *geometry-only* build: no HU-calibrated colony CT is available yet, so the pillar is bootstrapped on an *uncalibrated* museum microCT (*Saimiri sciureus* USNM 338948, UT Austin / DigiMorph, 0.1189 mm isotropic) registered to the VALiDATE29 multi-channel MRI atlas of the squirrel-monkey brain (Schilling *et al.* 2017; 29 animals, histologically-defined cortical labels). The atlas is staged and validated here (212<sup>3</sup> at 0.3 mm; its cortical parcellation is confirmed to be *region-level* — S1 = anterior parietal cortex, M1 = primary motor cortex, l/r separate); only the museum skull remains to be staged, so the cavity/registration outputs are the pending items. The pillar reuses TUBA's core primitives from the mouse/rat/macaque pillars almost unchanged — bone-shell cavity extraction (`tuba.core.cavity.bone_shell_cavity`), `cavity_binary` ANTs SyN (affine + deformable), atlas-into-subject warp, and the beam-aligned ( $c, \rho$ ) slab export. Its *distinctive* contribution is the **CT-calibration guard** for deliverable 1: a new module `tuba.core.hu_acoustics` implements the Aubry (2003) porosity  $HU \rightarrow (\rho, c, \alpha)$  model but *refuses to run* on input not flagged Hounsfield-calibrated (`UncalibratedInputError`), so a museum microCT can never silently produce quantitatively wrong acoustics; an explicitly-flagged placeholder ramp is the sanctioned geometry-only fallback. Deliverable 1 also reports the points-per-wavelength (PPW) budget across the 2–4 MHz operating band on the simulation grid, and a grid-convergence run of the transcranial time-of-flight aberration (the scalar a time-reversal correction must undo). All constants that can only be fixed against the *staged* scan (orientation flips, intensity thresholds, cavity hull geometry) are marked **PROVISIONAL**; quantitative cavity/registration results and QC figures are produced by the end-to-end driver once the scan is staged. The downstream wave physics (deliverables 3–7: time-reversal, multifocal synthesis, the two-foci feasibility sweep, MI/bioheat safety, and the per-animal export artifact) is out of scope for this branch and consumes the ( $c, \rho, dz$ ) slab in the `fullwave2-*` solver siblings. Code: `src/tuba/species/saimiri.py`, `tuba.core.hu_acoustics`, `tuba.atlases.validate29`; data manifest + fetchers at `src/tuba/data/`.

## 1 Scope and status

This is a *design and setup* report, not a validation report: the four prior TUBA pillars (mouse, macaque, human, rat) each document a pipeline calibrated end-to-end against a staged scan with measured cavity volumes, centroids, and focal figures. The squirrel-monkey pillar is at an earlier stage by deliberate choice — the funded project targets a Vanderbilt colony, whose HU-calibrated CT is not yet in hand, so we build and verify the *machinery* on a public museum microCT for geometry, gating the acoustics behind a calibration guard (§4) so nothing downstream mistakes an uncalibrated skull for a quantitative one. Two decisions frame the branch:

1. **Subject data:** a public museum microCT (§2), geometry only. The  $HU \rightarrow$ acoustics step refuses to run on it.
2. **Scope:** deliverables 1–2 (CT $\rightarrow$ acoustic model + atlas registration + target export + QC). Deliverables 3–7 (the wave-solver study) are out of scope here and live in the `fullwave2-*` siblings (§7).

Where a quantity can only be measured on the staged scan we write **[pending staged scan]** rather than a fabricated number.

## 2 Source data

**Skull.** The skull volume is a microCT scan of a *Saimiri sciureus* cranium, museum specimen USNM 338948 (National Museum of Natural History), scanned at the University of Texas High-Resolution X-ray CT Facility (UTCT) and distributed on DigiMorph [1]. The reconstruction is 540 axial slices at 0.1189 mm isotropic over a 54.0 mm field of reconstruction — a dry cranium, no soft tissue. The same openVertebrate / oUTCT

specimens are mirrored on MorphoSource, the collection the TUBA macaque pillar [9] also draws from (Copes *et al.* 2016 [2]). Table 1 summarises the parameters.

**Table 1.** *Saimiri sciureus* USNM 338948 scan parameters (DigiMorph specimen page). The intensity axis is *uncalibrated* (no HU phantom in the field of view).

|                         |                                              |
|-------------------------|----------------------------------------------|
| Specimen                | <i>Saimiri sciureus</i> , USNM 338948 (NMNH) |
| Facility                | UT Austin High-Resolution X-ray CT (UTCT)    |
| Slices                  | 540 axial                                    |
| Voxel                   | 0.1189 mm isotropic                          |
| Field of reconstruction | 54.0 mm                                      |
| Calvarium thickness     | 1.0–1.5 mm ( $\sim 8$ –13 voxels)            |
| Intensity               | uncalibrated reconstruction counts           |

**Uncalibrated intensity — the central caveat.** Museum microCT carries no hydroxyapatite/HU phantom, so voxel values are arbitrary reconstruction counts, not Hounsfield units. Every prior TUBA pillar faces the same issue and handles it by anchoring a histogram-relative intensity  $\rightarrow (c, \rho)$  ramp on the cortical-bone mode [8, 9]: robust to detector scale, but explicitly *not* quantitatively calibrated. For the squirrel monkey we make that limitation a hard boundary rather than a footnote (§4): the scan is flagged `UNCALIBRATED_MICROCT` and quantitative acoustics are refused.

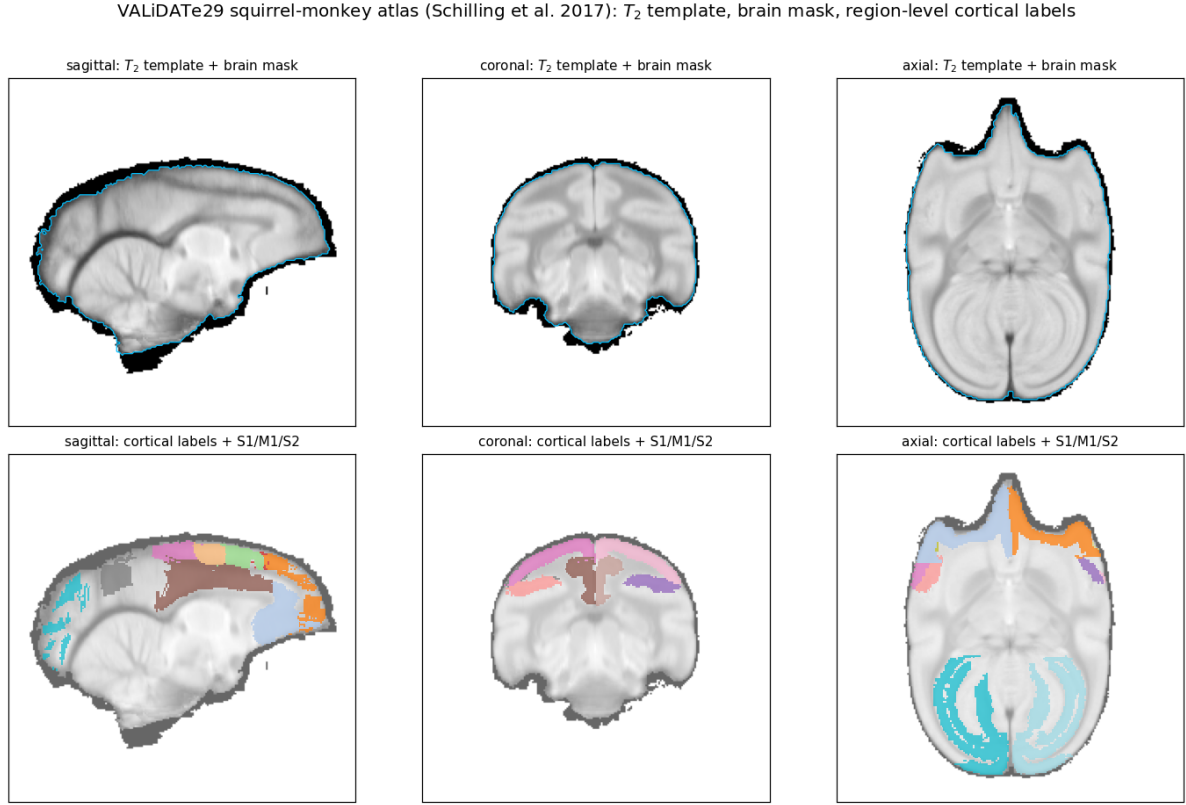
**Wavelength / PPW criterion.** The neuromodulation operating point is  $f_0 = 2$  MHz with a 2–4 MHz design range. The most restrictive wavelength is in the water column at 4 MHz:  $\lambda_{\min} = c_{\text{water}}/f = 1500/4 \times 10^6 = 0.375$  mm. A finite-difference solve wants  $\gtrsim 6$  points per wavelength there, so the simulation grid pitch is set to  $\Delta x = 60 \mu\text{m}$  (`SIM_DX_M` in `tuba.species.saimiri`); the resulting PPW budget is Table 2. Note the 0.1189 mm microCT is *coarser* than the  $60 \mu\text{m}$  sim grid, so the skull is up-sampled onto the sim grid by the slab sampler (§6) — the geometry is faithful; only the absolute acoustics are placeholder.

**Table 2.** Points-per-wavelength budget on the  $\Delta x = 60 \mu\text{m}$  simulation grid, using the shortest (water) wavelength at each frequency (`tuba.core.hu_acoustics.report_ppw`). All three bands clear the 6-PPW floor.

| Frequency | $\lambda_{\text{water}}$ (mm) | PPW @ $60 \mu\text{m}$ |
|-----------|-------------------------------|------------------------|
| 2.0 MHz   | 0.750                         | 12.50                  |
| 3.0 MHz   | 0.500                         | 8.33                   |
| 4.0 MHz   | 0.375                         | 6.25                   |

### 3 VALiDATE29 reference atlas

The brain anatomy is provided by the VALiDATE29 atlas of the squirrel monkey brain [3], built from 29 animals at the Vanderbilt University Institute of Imaging Science and distributed CC BY via NITRC (RRID:SCR\_015542). Unlike the subject skull, **the atlas is staged and validated in this report**: `tuba.data.fetch_validate29` discovers the current file-release link from the NITRC project page (reusing the WHS click-through of the rat pillar [10]) and unpacks it under `$VALIDATE29_DEST`; the fetch was exercised end-to-end here (v1.2 release, 30.4MB). Every volume is  $212^3$  at 0.300 mm isotropic. VALiDATE29 is multi-channel: in-vivo anatomical templates ( $T_2$ ,  $T_1$ , proton-density), diffusion templates (FA, MD), an ex-vivo structural + FA template, a *shipped brain mask* (`VALiDATE-brainmask`,  $\sim 33$  mL) and a discrete cortical + white-matter label volume (`VALiDATE3-labels`; 86 label ids) co-registered from Nissl/myelin sections. It supersedes the earlier single-subject squirrel-monkey MRI atlas [4]. Figure 1 shows the  $T_2$  template, brain mask, and cortical labels through the brain-mask centroid.



**Figure 1.** VALiDATE29 squirrel-monkey atlas through the brain-mask centroid (staged v1.2). **Top:**  $T_2$  template with the shipped brain-mask contour (blue). **Bottom:** region-level cortical labels (gray-matter ids 1–18) over the template, with the S1 (anterior parietal cortex), M1 (primary motor cortex), and S2 (PV/S2) target markers where the centroid slice intersects them. The squirrel monkey is near-lissencephalic, so the cortical areas form broad dorsolateral bands. Generated by `docs/saimiri/make_figures.py` from the staged atlas.

**Layout tolerance (a real calibration note).** The staged VALiDATE29.zip double-nests VALiDATE29/VALiDATE29/ and carries macOS `__MACOSX/.` resource-fork cruft, and its lookup table is a plain space-delimited id name ABBREV text file (not ITK-SNAP format) sitting beside a `LICENSE.txt`. The binding `tuba.atlases.validate29.VALiDATE29` walks the tree recursively, *ignores the cruft*, resolves the template/label/mask by keyword, prefers a label-named table over `LICENSE`, and raises with the candidate list rather than guessing — all confirmed against the staged files. The cavity-binary registration uses the shipped brain mask directly as its fixed image (no derive step, no ANTs). Paths are overridable by constructor argument or environment variable (`VALIDATE29_DEST`, `VALIDATE29_URL`).

**Target structures (confirmed against the staged LUT).** The study targets the *hand* representations in S1 (areas 3b, 1) and M1; S1 and M1 are only  $\sim 2\text{--}3\text{ mm}$  apart in the squirrel monkey, which is what makes the downstream two-foci feasibility question (§7) non-trivial. Inspecting the staged label table settles an important granularity point: **VALiDATE29’s histological parcellation is region-level, not Brodmann-area-level.** There is no separate “area 3b” / “area 1” / “area 4” label; the primary somatosensory hand areas 3a/3b/1/2 are bundled into a single *anterior parietal cortex* (APC) label and the primary motor cortex is one label, with left and right as *separate* label ids. The binding maps the study targets accordingly (S1  $\rightarrow$  APC, M1  $\rightarrow$  primary motor, S2  $\rightarrow$  PV) and is hemisphere-aware. The atlas-space centroids measured from the staged label volume (Table 3) are anatomically consistent: M1 sits rostral and slightly dorsal to S1 across the central sulcus, with clean L/R mirror symmetry. Splitting 3b from area 1, or isolating the hand knob within APC, needs a stereotaxic prior on the homunculus that the distributed label volume does not carry (§9).

**Table 3.** VALiDATE29 study-target labels and their *atlas-space* centroids (left hemisphere; right is the mirror), measured from the staged VALiDATE3-labels volume via `tuba.atlases.validate29`. Coordinates in the atlas RAS frame (mm).

| Target | Atlas label (l/r id)             | centroid ( $x, y, z$ ) | vox    | mL   |
|--------|----------------------------------|------------------------|--------|------|
| S1     | anterior_parietal_cortex (11/12) | (−10.1, 1.1, 10.3)     | 18 166 | 0.49 |
| M1     | primary_motor_cortex (3/4)       | (−7.4, 5.8, 12.0)      | 11 810 | 0.32 |
| S2     | parietal_ventral/S2 (7/8)        | (−13.5, 3.7, 4.6)      | 8 755  | 0.24 |

#### 4 Deliverable 1: CT → acoustic model, with a calibration guard

This is the pillar’s distinctive contribution and the reason a geometry-only build is safe. The module `tuba.core.hu_acoustics` provides a *quantitative* HU mapping and a hard guard around it.

**Porosity model.** For Hounsfield-calibrated input we use the Aubry (2003) [5] porosity model. Bone porosity is estimated from CT radiodensity,

$$\Phi(\text{HU}) = 1 - \text{clip}\left(\frac{\text{HU} - \text{HU}_{\text{water}}}{\text{HU}_{\text{bone}} - \text{HU}_{\text{water}}}, 0, 1\right), \quad (1)$$

with  $\Phi = 1$  water/marrow and  $\Phi = 0$  dense cortical bone, and the acoustic properties interpolate linearly in the bone fraction  $(1 - \Phi)$ :

$$\rho(\Phi) = \rho_{\text{water}} + (\rho_{\text{bone}} - \rho_{\text{water}})(1 - \Phi), \quad (2)$$

$$c(\Phi) = c_{\text{water}} + (c_{\text{bone}} - c_{\text{water}})(1 - \Phi), \quad (3)$$

with cortical endpoints  $\rho_{\text{bone}} = 1900 \text{ kg/m}^3$ ,  $c_{\text{bone}} = 2900 \text{ m/s}$  consistent with the rat/macaque pillars, and water  $\rho = 1000 \text{ kg/m}^3$ ,  $c = 1500 \text{ m/s}$ . Attenuation is taken proportional to bone fraction between a soft-tissue floor (0.5 dB/cm/MHz) and a cortical-bone value (8 dB/cm/MHz) [6, 7], reported in dB/cm/MHz and applied by the downstream solver. SI units throughout.

**The guard.** `AubryHUMapping.properties()` raises `UncalibratedInputError` unless the bound `CTCalibration.kind == 'hu'`. The squirrel-monkey scan is a museum microCT with no phantom, so the pillar binds `CT_CALIBRATION = UNCALIBRATED_MICROCT` and the quantitative path is refused by construction. Rather than silently emitting physically meaningless densities and sound speeds from arbitrary reconstruction counts, the caller must either (a) supply an HU-calibrated colony CT — one line, `CT_CALIBRATION = CTRCalibration(kind='hu', hu_bone_max=<scan>)` — which unlocks the quantitative mapping automatically, or (b) opt in explicitly to `placeholder_ramp()`, an intensity → ( $c, \rho$ ) ramp carrying nominal cortical endpoints and a `.placeholder = True` tag so downstream code can detect the uncalibrated origin. The slab loader (§6) uses the flagged placeholder ramp: geometry is faithful, absolute ( $c, \rho$ ) are nominal.

**PPW report + grid convergence.** `acoustic_model_report()` prints the calibration state, the PPW table (Table 2) at 2/3/4 MHz on the sim grid, and — given a placement and a staged scan — one grid-convergence run of the transcranial *time-of-flight aberration*

$$\tau = \int \left( \frac{1}{c(x)} - \frac{1}{c_{\text{ref}}} \right) dx, \quad (4)$$

the phase-insertion delay that a time-reversal / phase-conjugation correction must undo, computed along the central beam ray at  $\Delta x$  and  $\Delta x/2$  and reported as an absolute and relative change. This is a convergence of the *acoustic-property* model with voxel pitch, not of a full-wave solve (which lives in the solver siblings, §7); it is exactly the quantity the downstream aberration correction consumes, so it is the right model-side convergence to report here. The units and sign are checked in the offline test suite (homogeneous medium  $\Rightarrow \tau = 0$ ; a faster bone slab  $\Rightarrow \tau < 0$ ; a smooth profile converges as  $\Delta x \rightarrow 0$ ).

#### 5 Deliverable 2: geometry — cavity, registration, targets

The geometry stages reuse the rat/macaque dry-skull spine with no algorithmic change; only the species constants differ, and the ones that depend on the scan are `PROVISIONAL` until it is staged.

**Ingest + orientation (PROVISIONAL).** The DigiMorph TIFF stack carries no DICOM affine; as in the rat pillar [10] the storage→anatomy mapping is fixed by an orientation probe (per-slice bone-mass profile + mid-stack orthoslices). The defaults in `tuba.species.saimiri` (`SWAP_ROW_COL=True`, `VOXEL_SIGNS=(-1, -1, +1)`, storage (slice=*AP*, row=*DV*, col=*LR*)) follow the macaque/rat museum-CT convention and are *provisional* pending that probe. At 0.1189 mm the scan is already near the rat working-cache pitch, so a factor-1 reformat to RAS is used rather than a block-max downsample.

**Endocranial cavity (PROVISIONAL).** Squirrel-monkey whole-brain volume is  $\sim 22$  mL, so the cavity target is  $\sim 20$ – $28$  mL; the many dry-cranial openings (orbits, foramen magnum) rule out the mouse-style pure-hysteresis extractor for the same reason they do in rat and macaque. We therefore use the species-agnostic bone-shell extractor `tuba.core.cavity.bone_shell_cavity` (per-column basicranium floor + per-row plugs + per-axial hull), with hull half-extents interpolated between the rat ( $\sim 2$  mL) and macaque ( $\sim 90$  mL) pillars (`HULL_X_HALF_MM=18`, etc.). The produced volume is checked against the 20–28 mL target at run time and flagged if outside; the measured value is [\[pending staged scan\]](#).

**Registration — cavity\_binary SyN (affine + deformable).** Deliverable 2 calls for affine *and* deformable registration, so — unlike the macaque/rat pillars, which stop at a 12-DOF Affine — we use the full `cavity_binary` SyN path (`tuba.core.warp.register_subject_to_atlas`, fixed = VALiDATE29 brain mask, moving = subject cavity), the same convention the mouse and mini pillars use. The atlas template and cortical labels are then pulled back into subject space through the saved inverse transforms (`warp_atlas_into_subject`, linear for the template, nearest-neighbour for the labels). Unlike the rat and mouse pillars, which *derive* a brain mask from the annotation, VALiDATE29 ships its own (`VALiDATE-brainmask`), so the binding uses it directly as the SyN fixed image — no derive step and no ANTs for the mask.

**Target export + QC (deliverable 2 outputs).** `export_targets()` writes the S1 (anterior parietal cortex) and M1 (primary motor cortex) coordinates in the subject simulation frame to `saimiri_targets.json`, alongside the transform prefix, and `qc_figure()` writes the three-panel orthoslice overlay (bone shell, warped VALiDATE29 brain mask, S1/M1 markers). The transform is saved under `$TUBA_SAIMIRI_REG_DIR` with prefix `saimiri_cavity_to_validate29`. The target→label mapping and the *atlas-space* centroids (Table 3) are already confirmed against the staged atlas; only the *subject-space* coordinates and the QC figure remain [\[pending staged scan\]](#) the museum skull, since they are the warp of these labels through the cavity registration.

## 6 Placement and slab export

`place_on_skull` computes the apex/target/beam triple for a focused bowl aimed at a study target, and `load_slab` returns  $(c, \rho, dz_{\text{slab}}, \text{frame})$  with the *same signature* as the mouse/rat/macaque loaders, so any FDTD or angular-spectrum driver wired against one species runs on this one unchanged. The slab ramp is the flagged placeholder (§4); the cavity is overwritten with brain-parenchymal properties ( $c = 1540$  m/s,  $\rho = 1040$  kg/m<sup>3</sup>) exactly as in the other pillars. This slab is the hand-off boundary to the wave-solver siblings.

## 7 Out of scope here: the wave-physics study (deliverables 3–7)

TUBA is the geometry/atlas/targeting layer; the wave physics lives in the `fullwave2-*` FDTD siblings and consumes the  $(c, \rho, dz)$  slab and placement this pillar exports. The funded study’s remaining deliverables are therefore deliberately *not* in this branch:

- D3 time-reversal aberration correction through the individual skull (virtual source at each target → recorded element signals → phase-conjugate transmit delays);
- D4 multifocal beam synthesis — two simultaneous foci (S1 + M1) with independent amplitude control, plus at least one alternative to superposed time reversal (e.g. Gerchberg–Saxton);
- D5 the feasibility study (highest-priority result): sweep two-foci separation 1.5–4.0 mm at 2, 3, 4 MHz through the skull model, reporting –6 dB focal dimensions, peak-to-valley contrast, off-target ratio, and transcranial focal gain;
- D6 safety margins — MI at target and skull, and a bioheat estimate over the 400 kPa / 300 ms / 3 s duty cycle;
- D7 the per-animal targeting artifact (element delays, apodisations, target metadata, safety numbers).

The PPW budget (Table 2) and the time-of-flight grid-convergence (§4) are the model-side inputs D3 and D5 will build on.

## 8 Verification

Because no *skull* is staged, verification is by offline unit tests (`tests/test_saimiri.py`, 11 tests, no network / no antspyx / no data) — plus a live check of the atlas binding against the staged VALiDATE29 (keyword resolution, hemisphere-aware label lookup, shipped brain mask, and the target centroids of Table 3). The offline tests cover the logic that must hold regardless:

- importing the pillar stays ANTs-free (ANTs is lazy inside `tuba.core.warp` and the atlas binding);

- the calibration guard refuses uncalibrated input on every entry point, and the calibrated Aubry mapping is physically monotonic (porosity  $\downarrow$ ,  $\rho/c/\alpha \uparrow$  with HU) with the correct water / cortical endpoints and clamping;
- the placeholder ramp is flagged and maps to nominal endpoints;
- the PPW, time-of-flight-aberration, and grid-convergence maths (homogeneous  $\Rightarrow$  0; bone slab  $\Rightarrow$  analytic; smooth profile converges);
- VALiDATE29 file/label resolution and S1/M1 target keyword mapping on a synthetic atlas, plus a synthetic affine registration round-trip;
- the data manifest carries the three Saimiri entries with the expected calibration/license semantics.

The full repository suite is 42 passing (31 prior + 11 new); `ruff` is clean.

## 9 Open items

The atlas half is staged and validated (§3, Fig. 1, Table 3); the remaining items require the museum skull (and, for quantitative acoustics, an HU-calibrated colony CT):

1. Stage the museum microCT (`python -m tuba.data.fetch_saimiri` — interactive request/login), then run `python -m tuba.species.saimiri build`. The VALiDATE29 atlas fetch is already verified end-to-end.
2. Calibrate the PROVISIONAL constants (orientation flips, intensity thresholds, cavity hull) against the staged scan via the orientation/histogram probe, exactly as the rat pillar [10] documents; report the measured cavity volume against the 20–28 mL target and produce the cavity + registration QC figures, and warp Table 3 into subject space.
3. Add the stereotaxic hand-sub-representation prior within anterior parietal cortex (the label granularity limit of §3); the region→label mapping itself is now confirmed and pinned.
4. When a Vanderbilt colony CT is available, set `CT_CALIBRATION` to an hu calibration to unlock the quantitative Aubry mapping, and run the grid-convergence with the real property field.
5. Hand the exported slab + targets to the `fullwave2-*` siblings for deliverables 3–7 (§7).

## References

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